Methodological Criticism and Continued Citation

Ken Cor Gaurav Sood

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Abstract

Finding a methodological error can improve later research only if researchers respond to it. We examine citations after audits of statistical practice in neuroscience, animal experiments, and political science. Our neuroscience design compares flagged and unflagged papers published in the same journal and year, allowing each journal/publication-year group its own citation trajectory and accounting for stable differences between articles. OpenAlex histories through 2025 show continued citation: flagged papers' median annual citations rose from 6.5 in 2010 to 14–19 during 2012–2015 and were 7 in 2025. A separate citation-context sample recorded no acknowledgment of concerns in 94 of 95 valid completed ratings. Using historical Web of Science counts, the within-journal/year estimate of the proportional citation change is -8.5% (95% interval [-24.7, 11.2]%); rebuilding the same comparison with OpenAlex gives -7.3%. The estimate remains similar when we additionally compare studies of the same species and within/between-subject design. The additional audits also show continued citation. The findings document persistent attention to criticized research, usually without a recorded qualification in the neuroscience context sample, alongside an estimated relative citation penalty whose magnitude remains uncertain.

Keywords: methodological criticism; citation analysis; scientific self-correction; statistical errors; citation contexts

1 Introduction

Finding a consequential error in published research is only the beginning of correcting it. Subsequent researchers must learn about the problem and reconsider how they use the affected work. Otherwise, an unsupported inference can continue to motivate hypotheses, justify a method, or enter a synthesis of evidence. Whether public criticism changes later research therefore matters for the value of the effort spent identifying and explaining errors.

Studies of retracted research show why disclosure alone may be insufficient. Retracted articles continue to receive citations, and many citing passages do not mention the retraction (Hsiao and Schneider, 2021). Citation behavior can reflect how researchers search for relevant work: unfamiliarity with a field and reliance on prominent journals are associated with continued citation of retracted articles (Woo and Walsh, 2024). Criticism may also remain outside the citing literature. Articles discussed critically on post-publication review platforms do not necessarily attract corresponding negative citations in research papers (Bordignon, 2020). These findings make the transmission of corrective information an empirical question, even when the original article carries a formal warning.

Methodological criticism often supplies a less conspicuous warning. An audit can show that a statistical argument is invalid or that a result depends on a questionable analytical choice while the original article remains available in its published form. Researchers may have to connect a critique published elsewhere to the specific result they wish to cite. They may also reasonably continue to use portions of a paper that the criticism does not affect. The relevant questions are whether the affected papers lose citations relative to comparable papers and whether subsequent citations acknowledge the concern.

We examine those questions using four methodological audits. They concern an invalid comparison of statistical significance in neuroscience, pseudoreplication in animal experiments, weak instruments and sensitivity of inference in political science, and extrapolation in political-science interaction models. Each supplies paper-level assessments and a comparison group drawn from the same audit. These assessments range from identifiable errors in statistical reasoning to diagnostic concerns about an estimate. We preserve that distinction: a flagged diagnostic need not establish that the substantive finding is false, and an unflagged paper is not necessarily free of other problems.

Our outcome is citation to the original paper. The neuroscience design compares flagged and unflagged papers published in the same journal and year. These papers passed the same journals' editorial filters, come from the same assessment exercise, and have similar citation ages. Article fixed effects account for permanent differences in citation levels, while separate calendar-year effects for each journal/publication-year group account for different citation trajectories across those groups. We supplement this comparison with estimates from the other audits and an inverse-variance-weighted synthesis. We also examine citation levels and a historical sample of passages citing the neuroscience papers. Citation levels describe continued attention; comparative changes estimate how that attention differs from the trajectory of other assessed papers. A paper can remain widely cited and still receive fewer citations than it would have without criticism.

Figure 1 follows the neuroscience papers through 2025 in OpenAlex, showing persistent citation in both medians and means. The within-journal/year comparison estimates a -8.5% change in relative citation growth (95% interval [-24.7, 11.2]%). The point estimate is similar when citation histories are rebuilt from OpenAlex and when the sample is restricted to papers with a full calendar year after publication before the critique. The design therefore compares citation growth among papers of similar age in the same journals, rather than attributing ordinary differences between journals or publication cohorts to the critique.

Failed replications raise a related but separate question. Prior work has examined citations after unsuccessful attempts to reproduce findings, including the Reproducibility Project: Psychology (Open Science Collaboration, 2015; Serra-Garcia and Gneezy, 2021; Schafmeister, 2021; von Hippel, 2022; Clark et al., 2023). Our analysis concerns published methodological criticisms of the original research. An unsuccessful replication does not by itself identify an error in the original analysis, so replication outcomes are not pooled with these audits.

2 Data and research design

2.1 Audits, assessments, and publicity

We assemble audits for which the original papers, methodological assessments, and publication histories can be recovered. This is a selected collection of audits, not a probability sample of scientific errors or a systematic review of every published critique. Within each audit, we preserve the source roster and classifications, document missing or unclear assessments, and distinguish the papers available for description from those contributing to estimation. Source files, identity decisions, publicity dates and numbered acquisition and analysis scripts accompany each study in the repository.

The neuroscience audit reviewed behavioral, systems, and cognitive neuroscience papers published in 2009–2010 in *Nature*, *Science*, *Nature Neuroscience*, *Neuron*, and sampled issues of the *Journal of Neuroscience* (Nieuwenhuis et al., 2011). It identified 157 papers in which a comparison of effects was relevant and classified 79 as making the mistake. The mistake is to interpret statistical significance in one group and nonsignificance in another as evidence that the groups differ. Establishing that difference requires testing the contrast itself (Gelman and Stern, 2006). The audit's authors supplied us with the paper-level classifications. Although their article did not list the affected papers, a reader who understood the critique could recognize the mistake in an original article. We use the August 2011 publication as the publicity episode.

The animal-study audit assessed 200 experiments in which interventions were applied to parents and outcomes measured in offspring (Lazic et al., 2018). It classified 91 as using pseudoreplication, 45 as correctly analyzed, and 64 as too unclear to judge. Pseudoreplication treats dependent observations, such as offspring from the same treated litter, as independent experimental replications. The paper-level dataset and preprint became public in September 2017, before journal publication in April 2018. We exclude unclear assessments and one flagged paper criticized for the same problem in 2013. The descriptive comparison retains 90 flagged and 45 comparison papers.

The instrumental-variable audit reanalyzed designs in 67 political-science papers (Lal et al., 2024). Our primary diagnostic flags a paper if any assessed design has an effective first-stage F below 10. This is a screening threshold, not a universal definition of weak instruments. An alternative flags estimates significant under the archived analytic test but not under an Anderson–Rubin or applicable tF procedure. That comparison is restricted to papers with at least one analytically significant estimate. These definitions capture different methodological concerns and identify overlapping groups. The audit appeared online in May 2024; an earlier manuscript is dated July 2021. We therefore study the additional publicity surrounding formal publication.

The interaction-model audit flags 14 papers for severe extrapolation and leaves 8 unflagged on that diagnostic (Hainmueller et al., 2019). The concern is that estimated conditional effects rely on combinations of predictors with little supporting data. Comparison papers may have other problems identified by the audit. Online journal publication occurred in December 2018, following earlier draft circulation and a July 2018 replication archive. Here too, formal publication is an episode of additional publicity, not an established first disclosure.

2.2 Citation histories and citation contexts

We count distinct citing documents per original paper and year, retaining citation-free years when the history has been collected. Missing histories remain missing. Historical Web of Science exports cover 155 of the 157 neuroscience papers. Two covered histories contain the intended original itself and records predating its publication, indicating invalid searches. Excluding both leaves 76 flagged and 77 comparison papers. The incomplete 2016 Web of Science export is excluded. OpenAlex histories extend the descriptive series for the same papers through 2025, counting citing articles and reviews. Figure D1 in the appendix checks the descriptive pattern against the historical Web of Science data for 2009–2015. Appendix A documents the exclusions and record-level checks.

OpenAlex supplies the instrumental-variable and interaction-audit histories; their primary counts include articles and reviews. OpenCitations supplies the animal-study histories, including all indexed document types. The databases and available document-type filters differ, so the synthesis combines comparisons within each audit rather than treating their citation levels as directly interchangeable. We also reconstruct the neuroscience histories in OpenAlex and OpenCitations to compare estimates on identical original papers and years (Appendix D).

The historical citation-context sample concerns citations to flagged neuroscience papers after the critique. It contains 95 valid completed ratings. The protocol asks whether the citing text notes a concern about the cited paper; it does not require the concern to be the specific statistical error. Known false links, unavailable articles and unfinished ratings remain separately identified. The sample has no pre-critique coding, so it describes recorded acknowledgment after publicity rather than a change in acknowledgment. We do not infer the citing authors' awareness from these codes.

2.3 Comparisons and estimand

The target is the change in citations caused by publicizing a methodological problem. We estimate proportional citation changes for flagged papers relative to comparison papers, conditioning on the

comparison groups specified below. A negative coefficient means that flagged papers' citations grew less, or fell more, proportionally. It does not require their citation counts to fall in absolute terms.

For each audit, we estimate a Poisson pseudo-maximum-likelihood model with article and calendar-year fixed effects:

$$E[C_{it} \mid i, t, F_i] = \exp\{\alpha_i + \lambda_t + \beta F_i P_t\},$$

where C_{it} is the citation count, F_i denotes a flagged paper, and P_t denotes the post-publicity period. Article effects absorb stable differences in citation levels; year effects absorb common calendar-year changes. We report $100(e^{\widehat{\beta}} - 1)$ with article-clustered uncertainty. This approach retains observed zeros and does not require citation counts to have equal conditional mean and variance (Santos Silva and Tenreyro, 2006). Papers with zero citations in every included year do not identify the coefficient and are recorded separately.

For the neuroscience comparison, we replace λ_t with $\lambda_{s(i)t}$, where $s(i)$ is the article's journal and publication year. Each of the ten journal/publication-year groups contains both flagged and unflagged papers; all 153 originals therefore have comparison support. The design allows, for example, papers published in *Neuron* in 2009 to age differently from papers published in *Science* in 2010. It identifies the coefficient from differential changes within these groups. We use 2010 as the baseline and 2012–2015 as the follow-up, excluding the 2011 transition year. The identifying assumption is that, absent the critique, flagged and unflagged papers within a group would have followed comparable proportional citation paths. We assess comparability using the original audit's species and study-design classifications and pre-critique citations, and estimate additional comparisons within those study types. An additive fixed-effects model estimates the corresponding difference in citation-count changes. Both models impose a common within-group contrast; they are not equal-weight averages of heterogeneous article-level effects.

The synthesis applies the common article-and-year specification to each audit and compares the calendar year before the selected publicity year with the calendar year after it, excluding the publicity year. The windows are 2010–2012 for neuroscience, 2016–2018 for animal studies, 2023–2025 for instrumental variables, and 2017–2019 for interaction models. We combine one log contrast per audit using inverse estimated sampling-variance weights. A paper shared by the two political-science audits is omitted from the interaction component to avoid counting the same original twice. The main synthesis contains 367 contributing papers. It summarizes these audits with their chosen definitions and windows; it is not a universal effect of discovering an error.

A causal interpretation requires the flagged and comparison groups to have followed comparable proportional citation trajectories without the publicity episode. Membership in an audit helps align the research area and assessment process, but error status was not randomized. Topics, prominence, citation age and prior criticism may differ across groups. Publicity can also affect comparison papers, so the contrast measures the differential response to being flagged rather than the audit's total effect on scientific activity. Earlier circulation limits interpretation as an effect of first disclosure. We examine longer windows, earlier trends, journal and publication-cohort adjustments, alternative diagnostics, and paper-level influence in the appendices.

3 Continued citation and acknowledgment

3.1 Neuroscience citation trajectories

Continued citation extends beyond a few prominent papers and persists long after the critique. In OpenAlex, flagged papers' median annual citations rose from 6.5 in 2010 to between 14 and 19 during 2012–2015 (Figure 1). Both groups' citation counts generally declined after peaking around 2013. In 2025, flagged papers still received a median of 7 citations and a mean of 12.2; comparison papers received a median of 5 and a mean of 9.0. The historical Web of Science series shows the same early rise in both groups (Figure D1).

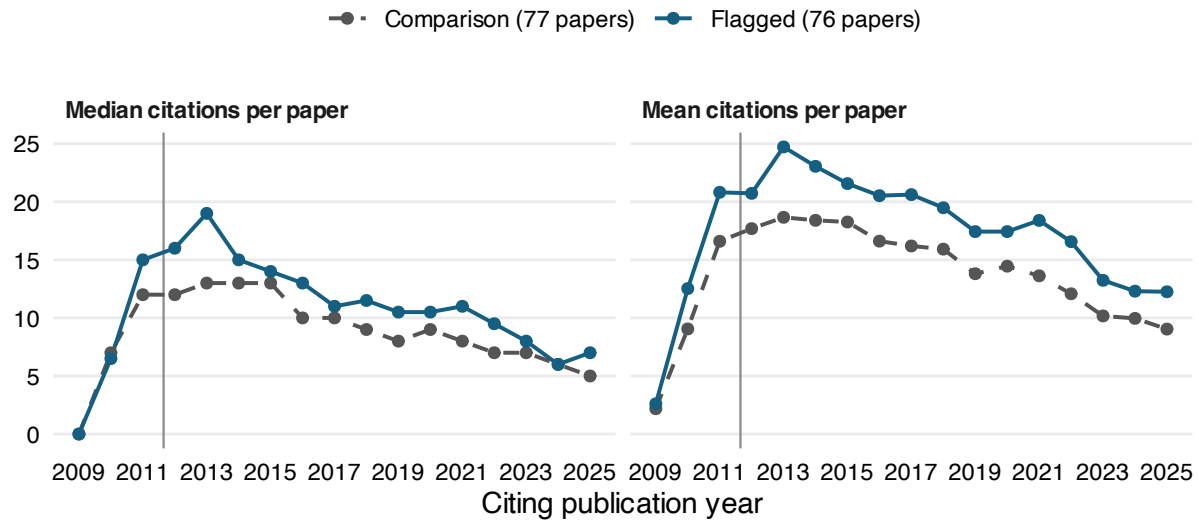


Figure 1: OpenAlex citation histories, 2009–2025. Median and mean annual citations from articles and reviews to the same 76 flagged and 77 comparison neuroscience papers in every year, including citation-free years. The vertical line marks the August 2011 critique. Duplicates and citations dated before the original paper are excluded. Early years include partial publication years and prepublication zeros for 2010 originals. The lines are descriptive; the regression compares 2010 with 2012–2015.

The within-journal/year regression uses historical Web of Science counts, compares 2010 with 2012–2015, and estimates a -8.5% change in relative citation growth (95% interval [-24.7, 11.2]%). OpenAlex gives -7.3% [-22.6, 10.9]% on the same papers and years. The estimates suggest slower proportional growth for flagged papers, even as their citation counts rose. The interval allows both a larger penalty and an increase, so the design does not tightly determine the magnitude.

An additive comparison that gives each flagged paper equal weight estimates 0.6 additional citations per paper per year (95% interval [-3.8, 5.0]). It compares each flagged paper's change with the average change of comparison papers in its journal/publication-year group. Flagged papers began with more citations, so they can gain more citations while growing less proportionally. Figure B1 in the appendix retains the median-and-mean presentation while standardizing both groups to the flagged papers' journal/publication-year composition. Appendix B reports support, timing,

source comparisons and the full-baseline-cohort sensitivity. The original specifications remain in Appendix C.

3.2 Acknowledgment in citing papers

Of 95 valid completed ratings, 94 recorded no acknowledgment of concerns and 1 recorded acknowledgment. Qualification was therefore rare in the observed post-publicity sample. The result concerns the neuroscience papers and cannot establish the frequency of qualification in the other audits.

A citation without criticism need not endorse the disputed inference. It may use another finding, a method, or background information. The codes also cannot distinguish failure to encounter the critique from a decision that the criticism does not affect the particular use. What they establish is that readers of these citing passages were seldom alerted to a concern about the original paper. Appendix A reports the coding protocol, missing ratings and uncertainty.

4 Citation changes across methodological audits

4.1 Citation levels and sensitivity within audits

The instrumental-variable papers continued to receive citations after formal publication of the critique. Under the two main diagnostics, flagged papers' mean annual citations rose from 7.0 to 7.2 and from 7.1 to 10.3 between 2023 and 2025; their medians changed little. Comparisons with other assessed papers are imprecise and change substantially when measured from an earlier year. The differing signs under alternative diagnostics do not establish distinct behavioral responses to different statistical problems (Appendix D).

In the animal-study audit, citation levels changed little on average. From 2016 to 2019, flagged papers' mean annual citations changed from 3.46 to 3.24, and comparison papers' from 4.96 to 4.82. Their respective medians were 3 and 2, versus 4 and 4. The difference in mean changes is -0.08 citations per paper (95% interval [-1.22, 1.07]). Appendix E reports longer trajectories and alternative comparisons. The synthesis uses the adjacent-year 2016–2018 contrast defined above.

The interaction-model comparison is sensitive to a highly cited comparison paper. Mean citations rose from 10.2 to 11.9 among flagged papers and from 10.4 to 14.5 among comparison papers. The medians moved differently: 8.5 to 10.0 and 5.5 to 3.0, respectively. The proportional contrast is -16.9% (95% interval [-43.4, 22.0]%); omitting the influential comparison paper changes it to 1.6%. Together with differences in earlier citation growth, this sensitivity limits interpreting the average contrast as a clear publication-related break (Appendix F).

4.2 Precision-weighted synthesis

Across the four audits, the main weighted contrast is -12.9% (95% interval [-24.3, 0.2]%). Table 1 reports the components. The point estimate indicates slower proportional citation growth among flagged papers. The interval includes little difference as well as a larger reduction. Replacing the instrument-strength diagnostic with inferential sensitivity gives -4.5% [-17.0, 9.9]%. This changes

the concern being assessed and the instrumental-variable comparison population; it is an alternative analysis of the same audit, not additional independent evidence.

Table 1: Relative citation growth across four methodological audits

Audit	Flagged / comparison	Change (%) [95% interval]	Weight (%)
Neuroscience	76 / 77	-14.1 [-32.8, 9.7]	33.6
Instrumental variables	8 / 59	-14.5 [-36.0, 14.2]	24.5
Animal studies	82 / 45	-7.2 [-29.4, 21.9]	27.0
Interaction models	13 / 7	-17.3 [-43.9, 22.0]	14.9
Precision-weighted mean		-12.9 [-24.3, 0.2]	100.0

Percentage contrasts compare flagged and comparison papers' post/pre citation ratios. The main instrumental-variable component uses effective $F < 10$. Counts describe papers contributing to the proportional estimate. All-zero papers remain in descriptive summaries, and the interaction component excludes the original already included in the instrumental-variable audit. Weights use estimated log-scale sampling variances. Component intervals use article-clustered standard errors and t_{G-1} critical values; the pooled interval uses a normal reference distribution. The synthesis excludes failed-replication studies and the small matched-case pilot.

The interval remains broad because article-level citation changes vary substantially and some comparisons contain few flagged or comparison papers. Highly cited papers can account for much of that variation. The synthesis uses only one pre-publicity year and one post-publicity year per audit, so it also retains annual fluctuations that longer histories could help distinguish from a sustained change. Its interval uses a normal critical value; the breadth does not arise from using a small-number-of-audits critical value. The repository provides an independent reconstruction of each standard error from paired citation counts.

The comparison assumptions pose a separate problem. In the interaction audit, slower relative growth precedes formal publication, and one comparison paper changes the sign of the post-publication estimate when omitted. In the instrumental-variable audit, changing the baseline year changes the estimates substantially. Both critiques circulated before journal publication. These findings weaken the interpretation of the pooled contrast as the effect of publicity. The reported interval does not account for bias from a wrong counterfactual or an incorrectly dated exposure. Appendix G specifies the weighting and reports random-effects uncertainty.

The original article-and-year neuroscience comparison also gives similar results across citation sources. On the same original papers and years, the relative-growth estimate is -11.5% in Web of Science and -10.9% using OpenAlex articles and reviews. A separate full-cohort OpenCitations comparison is also similar. This agreement concerns the estimated contrast, not completeness of the databases: the indexes differ in links and publication dates. Appendix D documents the paired comparisons.

5 Discussion

The affected papers continued to receive citations after public methodological criticism. The neuroscience evidence shows continued citation in medians as well as means, accompanied by little recorded acknowledgment in the citation-context sample. The additional audits show continuing

attention to papers flagged for other methodological problems. These observations address the persistence of the original research after criticism.

The comparative estimates address a different part of scientific correction: citations that criticism may have prevented. Their main precision-weighted point estimate indicates slower relative growth for flagged papers, but the magnitude changes with the diagnostic and comparison. Papers can remain influential while losing some of the citations they otherwise would have received. Our findings therefore do not support treating continued citation as proof that criticism has no effect, or treating a relative reduction as proof that the criticized inference has stopped circulating.

The distinction matters for how scientific criticism is communicated. A methodological article can teach readers to recognize an invalid inference, even without naming every affected paper. Making that connection still takes attention and judgment. Prominent citation histories do not show whether researchers have made it. The rarity of recorded qualifications shows that a concern is often absent from the citing text, while the citation counts reveal how much attention the original paper continues to receive. Neither measure alone establishes continued reliance on the particular disputed claim.

These audits also differ in what a reasonable response would entail. The neuroscience comparison of significance is an invalid argument for a difference between effects. Pseudoreplication misstates the amount of independent information in an experiment. Weak-instrument and extrapolation diagnostics instead question the reliability of particular estimates under particular assumptions. Such criticism can require a revised interpretation without making every use of the article inappropriate. We do not equate all flagged papers with false findings.

The within-journal/year design addresses differences in journals and citation age; its identifying assumption concerns remaining differences in growth within those groups. The additional audits have less uniform comparisons: some critiques circulated before journal publication, and earlier citation changes differ across diagnostic groups. Comparison papers may also respond to the critique. Our source cohorts are selected rather than representative of all methodological errors. The results describe the differential citation response around documented publicity episodes under explicit comparison assumptions. Establishing how criticism changes use of an erroneous claim would additionally require tracing that claim through later papers and observing whether the later argument actually depends on it.

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Data and code availability

Source inventories, paper identities, analysis panels, scripts, generated results and measurement documentation are available at `https://github.com/recite/sig_fault`. The repository preserves the original supplied classifications and coding records; the provider authorized redistribution of

the supplied neuroscience classification. Study-specific source and acquisition records distinguish public analytical inputs from cached bibliographic responses. The historical context ratings are archived observations, not newly collected or independently recoded judgments.

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A Data construction and citation-context records

The supplied classification spreadsheet identifies the articles by journal, publication year, page, and study characteristics. The identified Excel and CSV versions agree on the paper-level YES/NO codes. The citation import reads 16,437 database records across the covered sheets, repairs 1 malformed record, and marks 15 duplicate DOI records within cited papers. It removes 3 citation relationships identified as false in the historical context coding. Full source hashes and record-level indicators are generated by the replication scripts.

The unreliable histories are source IDs 23 and 25. Matching the intended papers' journal, year, and supplied page locates the target papers themselves in their supposed citing-article exports. Those histories also contain 54 records before the target publication year. This combination is consistent with exporting literature-search results instead of the intended citation results. The saved workbook contains no query log with which to reconstruct the precise operation. Deleting only the early records would not establish the validity or completeness of the remaining links. The principal estimates exclude both histories, while the appendix reports estimates retaining them. The other sheets contain neither a target-self match nor a record before the target year under these checks.

Missing sheets, source IDs 95 and 124, remain missing. In covered histories the completed 2010–2015 panel includes 29 paper-years with zero recorded citations. The scripts check that every paper has each required year and that joining the classification does not multiply or discard citation records.

Table A1: Disposition of the historical citation-context sample

Recorded status	Citation relationships
Article unavailable	1
Concern recorded	1
False citation link	3
No concern recorded	94
Uncoded	1

The sample consists of citation relationships from 2012 through the partial 2016 export. Every relationship matches a record in the source workbooks. "No concern recorded" and "Concern recorded" reproduce completed historical codes for valid citation relationships. The other categories preserve missingness and known false links.

The original coding protocol instructed the rater to assess whether a citing paper noted concerns about the cited paper, using the citing text and original article. The distributed file contains the rating and selected contextual text, not the full highlighted PDFs. We retain those ratings rather than claiming to have independently recoded the articles. The available materials contain one set of item-level ratings, so intercoder agreement cannot be independently checked.

Among valid completed ratings, the acknowledgment share is 1.1% (95% Wilson interval [0.2, 5.7]%). Assigning both unresolved, non-false records to either category gives a share between 1.0% and 3.1% among all non-false sampled relationships. These are missing-rating bounds, not sampling intervals. The Wilson interval is a binomial reference interval for sampled citation relationships; repeated source papers, selection, and coding errors are not represented by it.

B Within-journal and publication-year comparisons

The ten comparison groups are the five audited journals crossed with the two original-publication years. All contain both flagged and unflagged papers; the smallest contains three flagged papers and one comparison paper. Article effects absorb fixed citation levels, and group-by-calendar-year effects allow each group its own path. The main estimation retains all 153 papers. Standard errors cluster by article with the same finite-sample convention as the other models.

The repository's journal/year analysis records the group counts, article weights, model samples and annual estimates. It reports both citation sources, both outcome scales, the full sample and the 2009 publication cohort, and the longer and first-year follow-ups. The 2009 cohort has a full calendar-year baseline after original publication. A separate 2009–2010 contrast uses a partial publication-year baseline and is reported as a timing diagnostic, not a clean test of prior parallel trends. Annual intervals are pointwise.

The study-characteristic balance check finds that 25.0% of flagged papers study humans, compared with 55.0% of comparison papers after journal/year standardization. We therefore also compare papers within human/animal and within/between-subject groups, retaining 68 flagged and 52 comparison papers. A stricter species-and-design comparison retains 49 and 46, respectively. Each sensitivity first repeats the journal/year model on the retained sample, so the additional adjustment is not confused with a change in population. Table B1 shows similar proportional estimates. Country labels and baseline citation levels are not exactly balanced; their full diagnostics accompany the data. These checks support the comparison against specific observed differences, rather than establishing random assignment.

The fixed-effects models and the equal-flagged-paper contrast have different weights. With this common-date, balanced design, the linear coefficient weights each group's additive contrast in proportion to $n_{1s}n_{0s}/(n_{1s} + n_{0s})$. Those weights are nonnegative. The equal-flagged contrast instead uses $n_{1s}/\sum_s n_{1s}$, allowing heterogeneous group effects while giving each flagged paper equal weight. Its interval uses HC3 uncertainty from the equivalent weighted article-change regression, with residual degrees of freedom. Some finer groups contain a single comparison paper, so this interval is a model-based approximation rather than exact within-group randomization inference. The repository records outcome weights, leverage and variance contributions. It also records Poisson information shares, which describe curvature of the fitted objective rather than robust sampling variance or an exact weighting of heterogeneous percentage effects.

C Models, sensitivity, and uncertainty

C.1 Proportional changes

The original article-and-year specification, retained for comparison with the within-journal/year design, is

$$E[C_{it} \mid i, t, F_i] = \exp\{\alpha_i + \lambda_t + \beta F_i P_t\},$$

where F_i denotes a flagged paper and P_t denotes 2012–2015. Article effects absorb stable differences in citation levels and year effects absorb common calendar-year changes. Poisson pseudo-maximum

Table B1: Comparisons within study types, separating sample and adjustment

Source	Comparison	Flagged / other	Change (%)	95% interval
WoS	Journal/year, full	76 / 77	-8.5	[-24.7, 11.2]
WoS	Journal/year, 120 papers	68 / 52	-7.4	[-26.9, 17.2]
WoS	Human/animal and design	68 / 52	-5.6	[-27.7, 23.3]
WoS	Journal/year, 95 papers	49 / 46	-5.5	[-27.0, 22.4]
WoS	Species and design	49 / 46	-7.2	[-30.3, 23.6]
OpenAlex	Journal/year, full	76 / 77	-7.3	[-22.6, 10.9]
OpenAlex	Journal/year, 120 papers	68 / 52	-8.1	[-26.7, 15.1]
OpenAlex	Human/animal and design	68 / 52	-6.5	[-27.5, 20.6]
OpenAlex	Journal/year, 95 papers	49 / 46	-4.7	[-25.0, 21.1]
OpenAlex	Species and design	49 / 46	-6.1	[-28.4, 23.3]

All models include article and comparison-group-by-calendar-year effects, using 2010 versus 2012–2015. The additional groups distinguish human/animal research or exact species, and within/between-subject design, within journal and publication year. Intervals cluster by article. WoS denotes Web of Science. The two restricted journal/year rows use exactly the same papers as the corresponding study-type rows.

Table B2: Citation changes within journal and publication-year groups

Source	Publication years	Scale	Estimate	95% interval
Web of Science	2009–2010	Percent	-8.5	[-24.7, 11.2]
Web of Science	2009–2010	Citations	0.9	[-2.7, 4.5]
Web of Science	2009	Percent	-8.8	[-25.5, 11.8]
Web of Science	2009	Citations	2.3	[-2.0, 6.5]
OpenAlex	2009–2010	Percent	-7.3	[-22.6, 10.9]
OpenAlex	2009–2010	Citations	0.0	[-3.6, 3.5]
OpenAlex	2009	Percent	-9.9	[-25.7, 9.2]
OpenAlex	2009	Citations	0.7	[-3.1, 4.6]

Baseline: 2010; follow-up: 2012–2015. Percentage estimates use Poisson models; count estimates use linear models and are citations per paper per year. Both include article effects and journal/publication-year-by-calendar-year effects, with article-clustered 95% intervals. The 2009 cohort has a full calendar-year baseline after publication.

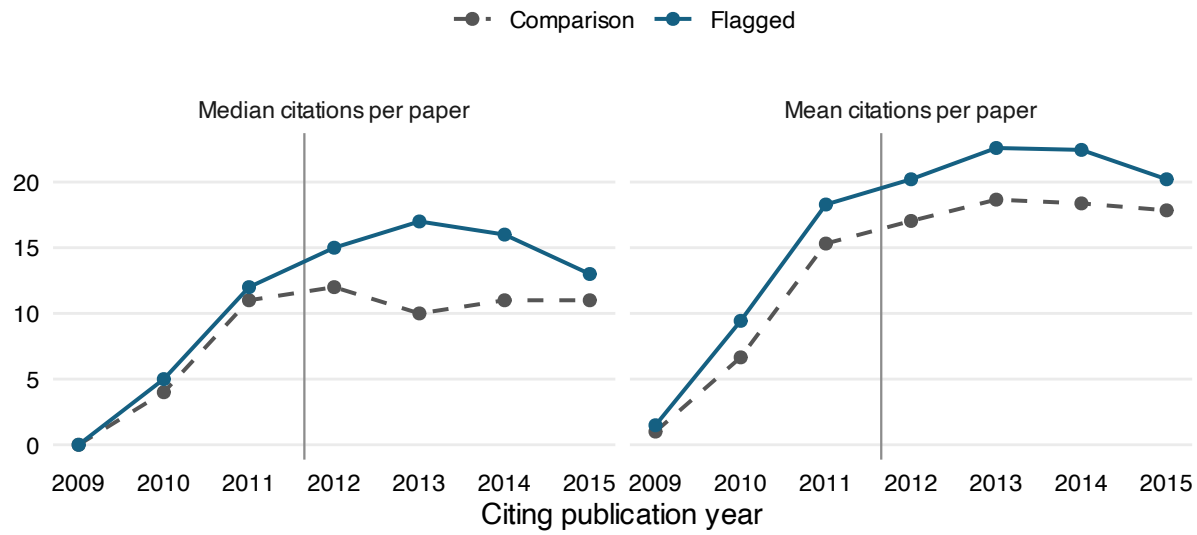


Figure B1: Historical Web of Science citation trajectories, 2009–2015, standardized to the flagged papers' journal/publication-year composition. Flagged papers have weight one; each comparison paper receives the ratio of flagged to comparison papers in its journal/publication-year group. The median is the first count reaching half of cumulative weight. The weighting uses no citation outcomes and differs from the regression's implicit weighting. The 2009 values include prepublication zeros for 2010 originals. Figure D1 shows the unweighted distributions from the same source and years.

likelihood retains zero counts and does not require equality of conditional mean and variance for the robust inference used here (Santos Silva and Tenreyro, 2006). The reported percentage is $100(e^{\beta} - 1)$, a relative post/pre citation change rather than a percentage change in the number of additional citations gained. In this balanced baseline specification, $e^{\widehat{\beta}}$ equals the ratio of the groups' post/pre mean citation ratios.

The covariance clusters score contributions by paper. We apply the $G/(G-1)$ and $(N-1)/(N-K)$ corrections, counting fixed effects not nested within article clusters in K, and use t_{G-1} critical values. Table C1 reports both two-sided 95% intervals and one-sided 95% lower bounds, transformed from the log scale. These are conditional model-based comparisons, not randomized-treatment intervals. No papers are excluded from this neuroscience follow-up fit for zero outcomes.

The article-and-year model's lower bound excludes proportional reductions greater than 28.7% at the one-sided 95% confidence level under this model. This leaves substantial reductions compatible with the data. The journal/cohort-adjusted lower bounds permit reductions of 22.0–28.5%; these bounds do not resolve uncertainty about the counterfactual trend. A stratified bootstrap resampling whole papers within each group gives a two-sided percentile interval of [-31.2, 14.5]% and a one-sided bound permitting a reduction of 28.3%. It uses 9,999 draws with seed 31415; no draws are discarded. The baseline ratio identity allows each bootstrap estimate to be calculated directly from the resampled paper-level before and after means.

Year-specific comparisons use 2010 as the baseline (Figure C1). A paper with zero citations in both the baseline and the comparison year has no finite Poisson article intercept and contributes no information to the slope; such papers are excluded from that fit and listed in the generated results.

Table C1: Proportional citation comparisons

Specification	Flagged / other	Change (%)	95% interval	Lower bound
Article and year effects	76 / 77	-11.5	[-31.6, 14.4]	-28.7
Journal-by-year effects	76 / 77	-11.4	[-31.4, 14.5]	-28.5
Cohort-by-year effects	76 / 77	-7.7	[-24.4, 12.7]	-22.0
Journal/year and cohort/year effects	76 / 77	-8.2	[-24.8, 12.1]	-22.3
Retain questionable histories	78 / 77	-12.2	[-31.9, 13.2]	-29.1
Retain duplicates	76 / 77	-11.6	[-31.6, 14.3]	-28.7
Retain false links	76 / 77	-11.5	[-31.6, 14.4]	-28.7
2009 publication cohort	44 / 47	-7.2	[-25.0, 14.9]	-22.3
2010 publication cohort	32 / 30	-10.4	[-49.8, 60.2]	-44.8
Post period: 2013–2015	76 / 77	-10.7	[-31.5, 16.4]	-28.5
Post period: 2014–2015	76 / 77	-11.7	[-32.7, 16.0]	-29.7
Potentially serious errors	43 / 77	12.8	[-18.3, 55.6]	-13.9

Change and bounds are percentages, $100(e^{\beta} - 1)$. The last column is the one-sided 95% lower bound. Every model includes article effects. The first row includes year effects; the next three replace common year effects with journal-by-year effects, publication-cohort-by-year effects, or both. Other rows use the first row's specification with the stated inclusion or post-period change. The baseline is always 2010. The potentially-serious subgroup retains the same comparison papers and excludes other flagged papers. Standard errors cluster by article.

The 2011 contrast describes the transition year. The available history is too short to justify fitting article-specific trends from predominantly post-publicity observations.

C.2 Absolute changes

For an absolute comparison, each paper's change is its mean annual citations in 2012–2015 minus its 2010 citations. Comparing the two groups' mean changes gives 1.8 (95% Welch interval [-2.1, 5.7]), allowing unequal variances of paper-level changes. OLS with article and year fixed effects on the same balanced panel reproduces this point estimate exactly; it is another implementation of the same comparison, not independent corroboration. The generated results also report its article-clustered interval. Journal and publication-year adjustment in the paper-level change regression uses HC3 uncertainty.

Removing each source paper in turn gives absolute point estimates from 1.3 to 2.5. A whole-paper stratified bootstrap gives a percentile interval of [-2.0, 5.6]. Medians describe typical citation levels; neither medians nor these sensitivity checks make error status or publicity exposure random. All analytical choices are retrospective.

Of the distinct citing documents in the main papers' 2010–2015 records, 13.7% cite more than one study paper. Shared citing documents and related topics can induce dependence across papers. Article clustering and the whole-paper bootstrap do not model those connections. The source review is also a selected set of journals and years, so these comparisons are not probability-sample estimates for all scientific publishing.

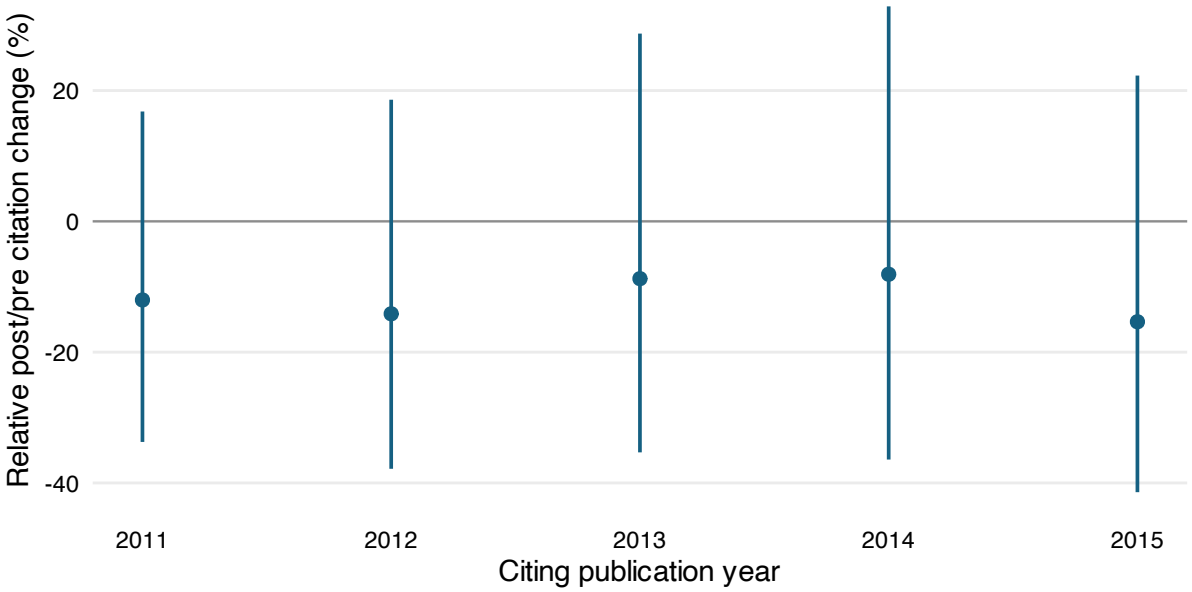


Figure C1: Year-specific proportional citation changes for flagged relative to comparison papers, using 2010 as the baseline. Points are $100(e^{\beta} - 1)$ from Poisson article/year fixed-effects models; bars are simultaneous 95% Bonferroni intervals across the five contrasts, with article-clustered uncertainty. Papers with zero counts in both compared years contribute no identifying information and are omitted from the corresponding fit. The 2011 contrast is transitional and is excluded from the main post period.

Table C2: Absolute citation changes under alternative comparisons

Comparison	Flagged / other	Difference	95% interval
Main comparison	76 / 77	1.8	[-2.1, 5.7]
Journal and publication-year adjustment	76 / 77	0.8	[-2.8, 4.4]
Retain both questionable citation histories	78 / 77	1.6	[-2.3, 5.5]
Retain duplicate database records	76 / 77	1.8	[-2.2, 5.7]
Retain known false citation links	76 / 77	1.8	[-2.1, 5.7]
Papers published in 2009	44 / 47	2.5	[-2.3, 7.4]
Papers published in 2010	32 / 30	0.3	[-6.4, 7.0]
Post period: 2013-2015	76 / 77	2.0	[-2.2, 6.2]
Post period: 2014-2015	76 / 77	1.7	[-2.5, 6.0]
Potentially serious errors only	43 / 77	4.3	[-0.1, 8.7]
Pre-critique change: 2009 cohort, 2009 to 2010	44 / 47	4.3	[0.1, 8.6]

Differences are flagged minus comparison, in citations per paper per year. Except for the final row, the baseline is 2010; the post period is 2012–2015 unless otherwise stated. Intervals are pointwise 95% Welch intervals, except the journal/year-adjusted estimate, which uses OLS with HC3 standard errors and a residual-degrees-of-freedom t critical value. The final row compares 2010 with 2009 among papers published in 2009; it is a diagnostic, not a publicity-effect estimate.

Table D1: Alternative diagnostics in the instrumental-variable audit

Diagnostic	Flagged	Comparison	Change (%)	95% interval
Effective F below 10	8	59	-14.5	[-36.0, 14.2]
Inferential sensitivity	11	44	23.9	[-7.1, 65.3]
Either diagnostic	17	50	11.8	[-14.6, 46.3]
AR-only sensitivity	3	52	-44.0	[-62.6, -16.0]

Change is $100(e^{\widehat{\beta}} - 1)$, comparing flagged and comparison papers' 2025/2023 citation ratios. Intervals use article-clustered covariance and t_{G-1} critical values. Inferential sensitivity and the Anderson–Rubin-only comparison restrict to papers with an analytically significant estimate. These overlapping definitions are alternatives, not independent studies. Only 3 papers meet the narrower Anderson–Rubin diagnostic.

D Additional-cohort construction and comparability

The IV replication archive contains multiple designs for some papers. We retain the design records but assign each paper one citation history and equal weight within its diagnostic group. The repository reproduces the point estimates for the 10 designs selected in the separate citation-context pilot. For the Alt design, the instrument is an eight-level factor with seven excluded indicators. Recomputing the joint Anderson–Rubin test with that encoding corrects the archived calculation; its loss-of-significance classification remains the same. The single-instrument tF calculation is inapplicable and does not enter that design's sensitivity classification. These checks do not constitute independent replication of every diagnostic in the full audit.

The relative citation-growth estimates are -14.5% for the weak-instrument screen and 23.9% for inferential sensitivity. On a fixed older cohort, moving the baseline from 2023 to 2022 changes the weak-instrument estimate from -11.5% to -0.4%, and the sensitivity estimate from 15.7% to -17.0%. Holding the cohort fixed shows that changing the starting year alone can substantially change the comparison.

Document types materially affect the IV comparison. The original broad outcome included preprints, proceedings and book chapters; some books contributed links from numerous chapters and front matter to the same original paper. After observing this problem and the alternative estimates, we selected articles and reviews as the primary outcome. For the weak-instrument screen, the broader count gives 39.3%, compared with -14.5% for articles and reviews. This was an outcome-aware measurement revision. The repository retains the broader results and a partial sensitivity that collapses identified Oxford book DOI families; neither procedure resolves all cross-version duplication.

D.1 Citation-source robustness

Figure D1 repeats the descriptive comparison in Figure 1 using the historical Web of Science exports. Over the shared 2009–2015 period, both sources show an early rise in citations to flagged and comparison papers. In Web of Science, flagged papers' median annual citations rose from 5 in 2010 to 13–17 during 2012–2015; comparison papers' medians rose from 4 to 10–11. Mean citations rose from 9.4 to a post-period annual average of 21.4 for flagged papers, and from 6.5 to 16.7 for comparison papers. For 91% of flagged papers, the post-period annual average exceeded the

baseline. This checks the early descriptive pattern across citation sources; the historical exports do not cover the later OpenAlex follow-up.

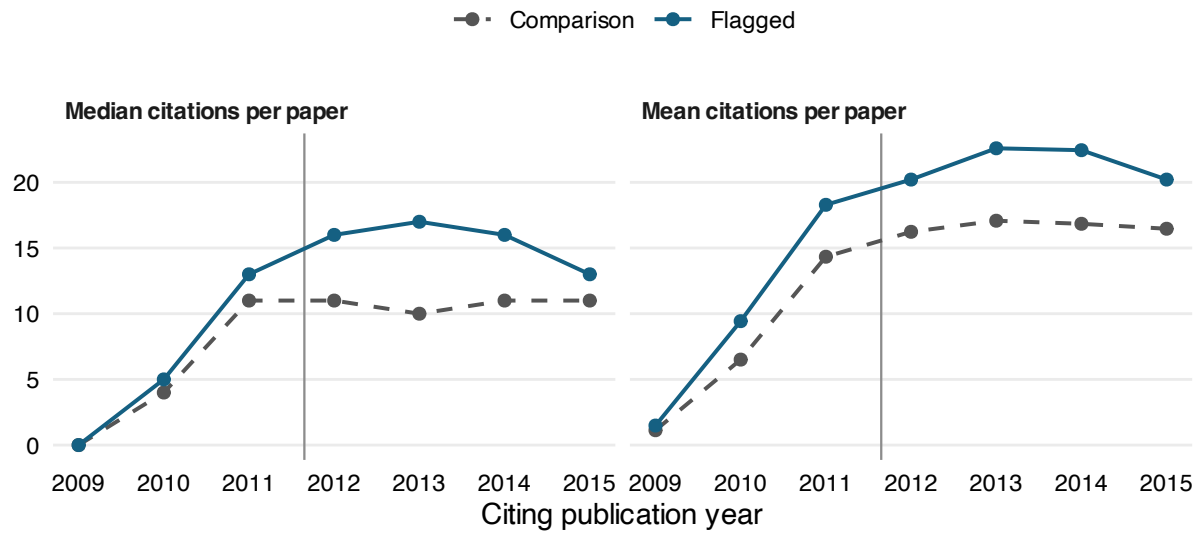


Figure D1: Citation-source robustness: historical Web of Science histories, 2009–2015. Median and mean annual citations to the same 76 flagged and 77 comparison neuroscience papers shown in Figure 1, including citation-free years. The vertical line marks the August 2011 critique. Early years include partial publication years and prepublication zeros for 2010 originals. The incomplete 2016 export is excluded.

The source-comparison protocol preserves the original neuroscience analysis cohort and compares identical papers and years before considering a pooled result. Exact DOI matches identify shared citing relationships, with each database's publication year retained. Historical exports lack the detailed document-type field needed to impose an identical article/review filter. Missing histories stay missing; a relationship absent from one database is not automatically a false citation. Source discrepancies therefore measure differences between the recorded citation histories rather than bias relative to an assumed ground truth.

The complete paired comparison contains 76 flagged and 77 comparison papers. Replacing the historical counts with OpenAlex articles and reviews gives a relative-growth contrast of -10.9% (95% interval [-29.0, 11.9]%), compared with -11.5% in the historical exports. The paired, paper-level bootstrap estimates a source-induced change in the growth ratio of 0.8% (95% interval [-8.9, 12.4]%). The broader OpenAlex document-type count gives a contrast of -10.7%. These are comparisons of recorded citation histories on fixed papers and years; neither database is assumed to be ground truth.

Among 3,223 shared citation relationships with conflicting years, OpenAlex dates 3,219 earlier. Publisher-deposited online and print dates explain 151 differences in the reviewed links; the remaining dates have not all been independently reviewed. Moving shared eligible citations between years contributes 242 citations to the 431-citation difference in the 2010 totals. Coverage and publication dating can therefore both change measured growth.

For the 153 paired papers checked against OpenCitations, that index also records 2,141 of the 2,637 OpenAlex-only links eligible for the primary article/review count. Across all document types in the comparison frame, it records 3,451 of 4,719 OpenAlex-only links. This corroborates many additional records, but shared upstream sources mean agreement across indexes is not independent verification of the citing bibliography. Undated OpenCitations links remain undated, and this check does not replace either source's annual counts.

A separate OpenCitations comparison covers all 153 historical papers. Its estimated relative-growth contrast is -11.9%, compared with -11.5% in the historical exports. OpenCitations records more citations in both periods, leaving this aggregate comparison almost unchanged. The outcome counts dated citing works across document types; 310 undated relationships remain unassigned. A paired, paper-level bootstrap gives a source-induced change in the estimated growth ratio of -0.4% (95% interval [-8.0, 9.2]%). These intervals condition on the observed cohort and database records. Similar aggregate estimates do not verify individual links or establish either source as ground truth.

For comparison across audits, the repository also reports the contrast between the calendar year before and the calendar year after the selected publicity year, excluding that year itself. This uses 2010 and 2012 for neuroscience and 2023 and 2025 for the IV audit. The neuroscience papers published in 2010 have a partial publication-year baseline, so a separate 2009-publication cohort has a complete baseline year. The synthesis below uses these common windows; alternative IV diagnostics are not counted as independent replications.

E Pseudoreplication audit: construction and sensitivity

The source sampled eligible English-language prenatal-exposure experiments in multiparous mammals from 500 PubMed search results. Our population is the 200 deposited assessments, not animal research generally. Exact PMID matching identifies all 200 papers; 198 also have a DOI. We retain the 2 papers without DOI and query their citations by PMID. The first preprint appeared on September 2, 2017, and the identified dataset was released on September 6. The repository documents the release dates and source provenance.

OpenCitations supplies 8,818 relationships. Citing records sharing a DOI or OpenCitations identity count once per target; 70 relationships have no assignable year. Counts include all indexed document types, including distinct book chapters. A check of 18 chapter–target pairs in three prominent book families verifies separate bibliographies containing the target references. We retain those chapters. A completed API history permits observed zero years; acquisition failure would not. The repository records identities, source checksums, citation links and notice adjudications.

One flagged paper received a public methodological criticism on May 17, 2013, with a response disputing the criticism. We exclude it from the first-known-publicity comparison and retain all 136 classified papers in a sensitivity. This eligibility change preceded estimation. Earlier errata for other papers include name and affiliation corrections; two notices remain incompletely reviewed. Excluding all papers with linked prior errata is an additional sensitivity, not a claim that every notice concerned the statistical problem.

The main absolute contrast is the difference in equal-paper mean changes, with a Welch interval. Article/year fixed-effects Poisson estimation gives a proportional contrast of -3.5% (95% interval [-25.4, 24.7]%). The 10 papers with zero citations in both selected years drop from that fit but remain in the absolute contrast and descriptive summaries. The full-cohort and prior-erratum exclusions, a 2019–2021 post-period average, and a fixed cohort published by 2013 yield similarly inconclusive comparisons. Split-unit designs are more common among flagged papers; standardizing within those design strata gives a difference of -0.57 citations (bootstrap interval [-1.61, 0.47]). The older cohort also had differing mean citation gains before publicity, weakening a causal interpretation. Full estimates and whole-paper bootstrap intervals are generated in the repository.

E.1 Synthetic difference-in-differences

Synthetic difference-in-differences gives a similar additive estimate to ordinary difference-in-differences on the same papers and years (Table E1). The synthetic estimate is -0.89 citations per paper per year (95% interval [-2.01, 0.22]), compared with -0.78 under ordinary DiD. Flagged papers averaged 2.89 citations annually in 2018–2020; the level-adjusted synthetic comparison predicts 3.79. These estimates concern the older papers with sufficient baseline histories, not the full audit population.

We use the estimator of Arkhangelsky et al. (2021), with nonnegative comparison-paper and baseline-year weights, article and time intercepts, and the authors' default regularization. The main comparison includes originals published by 2013, uses 2014–2016 as the baseline, and omits the 2017 disclosure year. Every flagged paper and follow-up year receives equal target weight. The interval uses 1,999 whole-article bootstrap draws, refitting both sets of weights while retaining the original regularization parameters. It allows serial dependence within articles and assumes independence across articles, conditional on this publicity event. A causal interpretation requires the adjusted synthetic trajectory to represent citation changes absent the disclosure.

The synthetic comparison uses the donors broadly: its effective donor count is 25.86 of 26, with a largest weight of 4.42%. Prediction before disclosure is less reassuring. A fresh fit trained on 2014–2015 predicts the flagged papers' 2016 citations with a gap of -0.36, compared with -0.20 under ordinary DiD. Comparison papers' 2016 counts enter the time weights, but flagged papers' 2016 counts and all later counts do not enter weight fitting. Synthetic weighting therefore does not improve this particular prediction check. An older cohort supplies a fourth baseline year; that year receives zero weight in the baseline adjustment, though it still enters donor fitting. The repository reports every donor and time weight, publication-cohort and study-design balance, sample disposition, and bootstrap draw. None of these alternative estimates enters the synthesis as an independent audit.

F Interaction-model audit: sample, timing and influence

The published audit's Table A1 records 46 interactions in 22 original papers. We preserve all interactions and aggregate each diagnostic separately: any adverse result flags a paper; if none is adverse but a test is missing, the paper remains unknown. Severe-extrapolation labels are complete.

Table E1: Citation changes after the pseudoreplication audit: synthetic and ordinary DiD

Sample/window	Flagged / controls	Synthetic DiD [95% CI]	Ordinary DiD [95% CI]
Main	50 / 26	-0.89 [-2.01, 0.22]	-0.78 [-1.86, 0.30]
Older, four pre-years	30 / 14	-0.82 [-2.13, 0.48]	-0.88 [-2.37, 0.62]
Older, three pre-years	30 / 14	-0.84 [-2.22, 0.54]	-0.55 [-1.99, 0.88]
Two post-years	50 / 26	-0.84 [-1.81, 0.14]	-0.66 [-1.72, 0.41]

Differences in annual citations per original paper, using distinct OpenCitations works of all types. Main baseline: 2014–2016; follow-up: 2018–2020. The older sample includes papers published by 2012; its four-year baseline starts in 2013. The shorter follow-up ends in 2019. Synthetic intervals use the article bootstrap; ordinary intervals use Welch uncertainty on article changes. The already documented 2013 warning is excluded throughout.

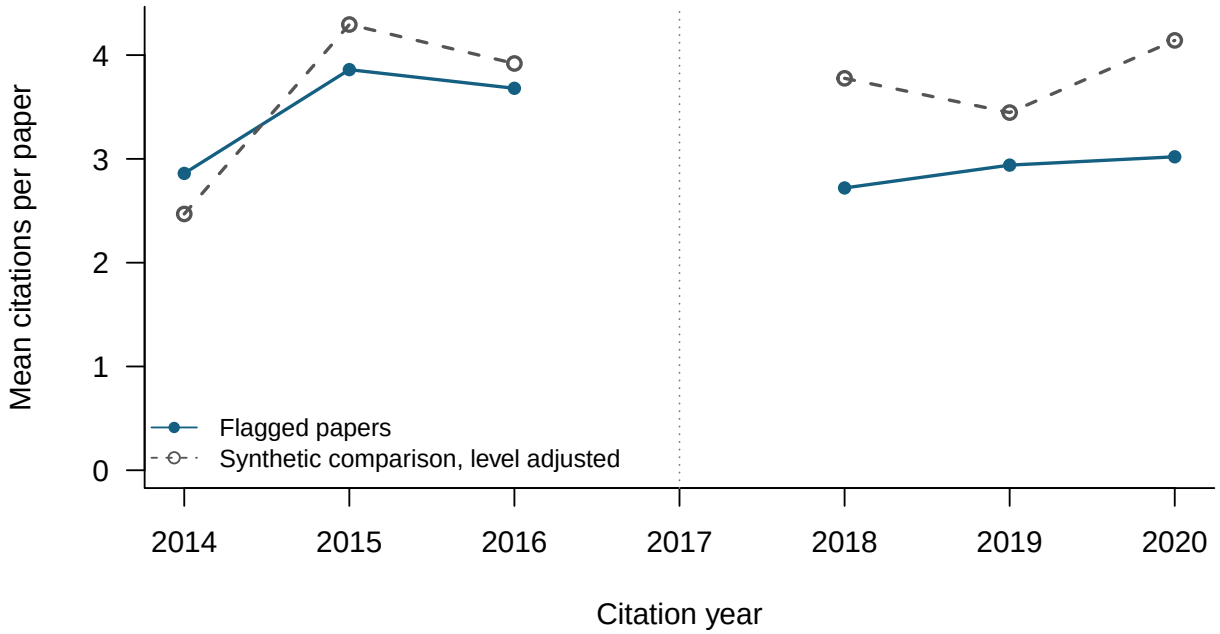


Figure E1: Mean annual citations to flagged papers and their synthetic comparison in the pseudoreplication audit. The comparison is shifted by the estimated weighted pre-period level difference. The disclosure year, 2017, is omitted. The difference between the mean post-period paths equals the synthetic DiD estimate; fitted pre-period proximity does not establish the absent-disclosure trajectory.

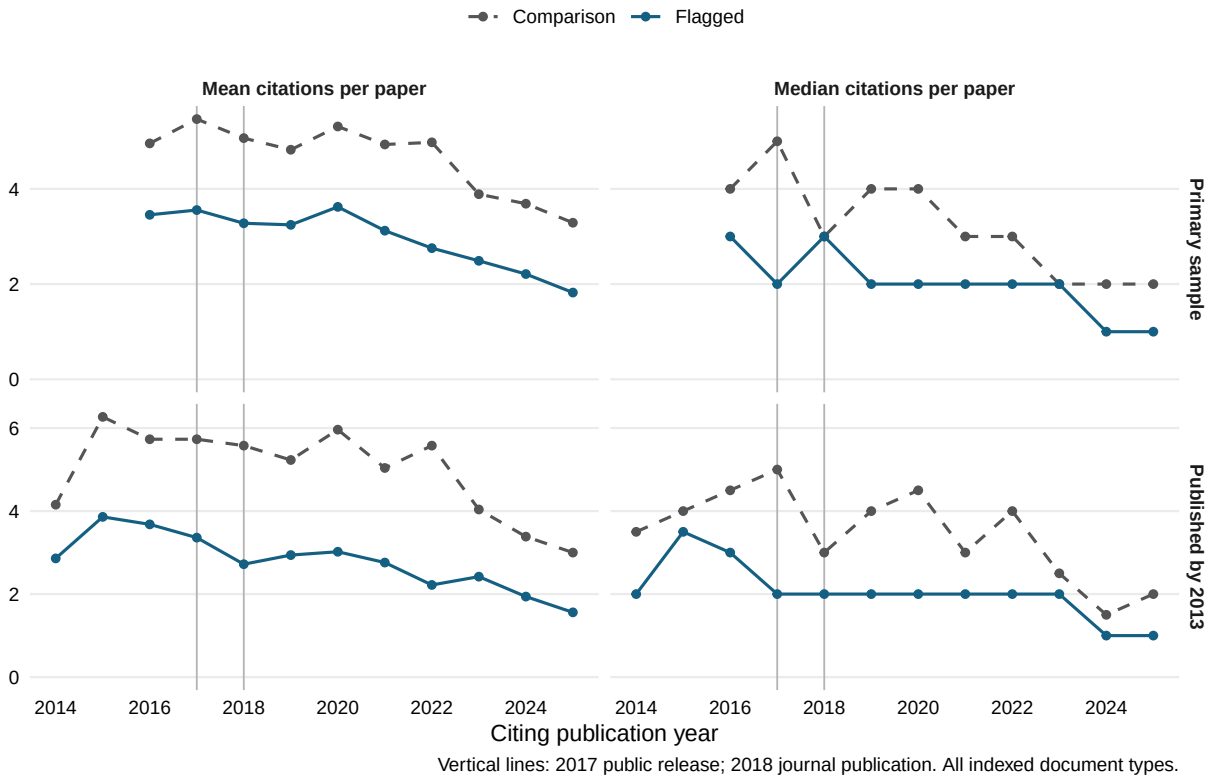


Figure E2: Mean and median citations in the animal-study audit. The primary sample contains 90 flagged and 45 comparison papers. The older fixed cohort contains 50 and 26, respectively. Vertical lines mark the 2017 public release and 2018 journal publication. Earlier-warning papers and unclear classifications are excluded. Counts include all indexed document types; citation-free covered years remain zero.

Of the 8 comparison papers, 6 have another adverse diagnostic. Neither this support diagnostic nor a failure to reject equal low/high effects establishes that the original substantive conclusion is false.

Publisher-deposited references identify every original, with author, print year and journal checked against publisher metadata. The version-one replication archive, publicly released on July 28, 2018, contains the 22-paper roster and analyses. It does not establish that all final binary labels appeared in the same form then. An SSRN draft was first posted in February 2016, but the earliest roster remains unverified. The event is therefore the December 18, 2018 online publication. The authors' December 2019 appendix update and response to subsequent methodological criticism fall within the longer follow-up sensitivity.

Complete OpenAlex citation links supply the annual panel. The primary outcome counts distinct citing articles and reviews. One paper, Williams (2011), has zero citations in both 2017 and 2019; it stays in descriptive means and medians but contributes no information to the Poisson coefficient. The standalone fit includes 21 contributing papers; excluding Vernby (2013) for the meta-analysis leaves 20. These exclusions follow the sample and estimation rules written before citation acquisition.

Article and year fixed-effects Poisson estimation yields the same coefficient as the log ratio of the two groups' post/pre mean citations. Article-clustered intervals use a t_{G-1} critical value, where G is the number of contributing papers. The within-group paired article bootstrap uses 9,999 draws and gives [-38.2, 41.9]%. The absolute difference between changes is -2.5 citations per paper per year (Welch interval [-10.6, 5.6]). Leave-one-paper-out proportional estimates range from -23.5% to 1.6%; the positive endpoint omits Huddy, Mason and Aarøe (2015). The negative earlier-trend contrast also cautions against interpreting the post-publication comparison as a newly emerging difference.

The repository reports 2019–2021 follow-up, all citing document types, separate linearity and low/high diagnostics, the 2015–2017 earlier trend, and exclusion of Malesky's previously criticized paper. Journal/year effects restrict to journals containing both groups: International Organization has only flagged papers. This leaves 19 originals, including the all-zero paper. These are sensitivity analyses of a small diagnostic-specific comparison, not independent studies to pool. The full source ledger, numbered scripts, receipt hashes and model results accompany the public annual panel.

G Synthesis across methodological audits

For each audit, we estimate the log of the flagged group's post/pre mean citation ratio divided by the comparison group's ratio. We weight these contrasts by inverse estimated sampling variance: $w_j = s_j^{-2} / \sum_k s_k^{-2}$ and $\bar{\beta} = \sum_j w_j \widehat{\beta}_j$. The reported quantity is $100(e^{\bar{\beta}} - 1)$, with standard error $1/\sqrt{\sum_j s_j^{-2}}$ on the log scale. Normal intervals condition on the included audits and estimated component variances. They summarize a precision-weighted average and assume independence across audit estimates. Weighting, timing, and selection of audits are retrospective choices.

The windows are 2010 versus 2012 for neuroscience, 2023 versus 2025 for instrumental variables, 2016 versus 2018 for animal studies, and 2017 versus 2019 for the interaction audit. These

are post/pre ratios, not annualized growth rates. For animal studies, 2018 follows the first public release but coincides with journal publication. Earlier circulation precedes formal publication of the instrumental-variable and interaction audits. The longer neuroscience and animal-study comparisons in the main text address persistence over their respective follow-up periods and do not replace these adjacent-year components.

Vernby (2013) appears in both the interaction and instrumental-variable audits. We omit it from the interaction component, while retaining it in the standalone interaction analysis. Papers with no citations in either comparison year contribute no information to the Poisson coefficient; they remain in descriptive summaries. The main synthesis contains 367 contributing original papers. No known original DOI overlaps the resulting components. Distinct originals may still share citing papers or correlated field shocks, which the component-wise sampling variances do not model.

The alternative instrumental-variable diagnostic replaces the instrument-strength component with the inferential-sensitivity component, including its restricted comparison population. It does not add an independent study. Citation-source substitutions and leave-one-audit-out estimates similarly change a comparison rather than increase the amount of independent evidence. The random-effects sensitivity uses restricted maximum likelihood and modified Knapp–Hartung intervals with three degrees of freedom. Its adjustment cannot shrink uncertainty below the unadjusted value. It estimates a modeled mean across audit effects; it is not a prediction for an unobserved critique. The result is -12.9% (95% interval [-30.7, 9.4]%).

Other source inventories do not enter until their identities, assessment definitions, publicity timing and citation comparisons have been established. Failed-replication citation analyses concern a different assessment and are not components of this synthesis.